JAY DARJI

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Machine Learning Engineer with **2+ years of ML/AI experience** building classification, ranking, retrieval, agentic, and inference systems from experimentation through APIs and evaluation. Strong in **Python, PyTorch, model evaluation, ML systems, and AI-assisted development**, with **3 peer-reviewed papers and 90+ citations**.

TECHNICAL SKILLS

Programming: Python, Java, SQL, Git/GitHub; data structures, algorithms, REST APIs, software testing
ML/AI: PyTorch, TensorFlow, scikit-learn, Transformers, classification, ranking, NLP, computer vision, time-series modeling
Retrieval and LLM Systems: Embeddings, semantic similarity, FAISS, RAG, prompt evaluation, LLM/agent workflows
ML Systems and Cloud: FastAPI, Docker, AWS Bedrock, MLflow, OpenTelemetry, SigNoz, model serving, GPU profiling
AI Development: Claude, OpenCode, AI-assisted coding, debugging, code review, and test generation

EXPERIENCE

Machine Learning Engineer Aug 2026 – Present
CHIPChop, Berkeley SkyDeck Startup (Stealth Mode)
- Built **AI agents on Amazon Bedrock** that generate, lint, simulate, and repair hardware code with restricted tool access.
- Built automated model evaluation using compilation, simulation, mutation testing, and **MLflow regression gates**.
- Instrumented agent workflows with **OpenTelemetry and MLflow tracing** for latency, failures, tool activity, and debugging.
- Developed reusable **Python simulation and grading services** with execution timeouts, typed results, and failure diagnosis.

Machine Learning Research Assistant Jun 2025 – Feb 2026
San Jose State University, San Jose
- Partially fine-tuned a pretrained vision transformer for 4-class MRI classification across 3 random seeds.
- Reached **98.83% ± 0.21% test accuracy and 98.97% ± 0.18% macro-F1** on 995 held-out images.
- Built **3 FastAPI inference services** and a modular workflow for prediction, explanations, and generated summaries.
- Used an independent LLM judge to check generated summaries for consistency and grounding.

Data Scientist Sep 2022 – May 2024
Rhenix Lifesciences LLP, Hyderabad
- Led 3 ML projects across forecasting, classification, missing-data modeling, and time-series feature engineering.
- Beat ARIMA on 10-subject CGM data: **10 mg/dL RMSE at 15 min and 17 mg/dL at 60 min**.
- Built a cross-validated Random Forest sleep-wake classifier on wearable data, reaching **93% accuracy and 0.96 ROC-AUC**.
- Developed a novel binning method that reduced imputation RMSE in **70–90% of tested windows**.
- Built ETL, feature-engineering, quality-control, and Tableau workflows for wearable time-series analysis.

Back-End Software Developer Dec 2021 – Apr 2022
Cognizant Technology Solutions Pvt. Ltd., Bengaluru
- Developed Java Spring Boot microservices and integrated **REST APIs** for client-server applications.
- Optimized MySQL queries with indexing and query tuning, reducing execution time by **20%**.

Machine Learning Research Intern Jun 2019 – Aug 2019
Institute of Infrastructure Technology Research and Management, Ahmedabad
- Co-authored an EOG/EMG sleep-disorder study; achieved **94.3% six-class accuracy** with 10-fold cross-validation.

PROJECTS

Candidate Recommendation Engine GitHub
- Built a resume-to-job **ranking system** with SBERT, OpenAI/Gemini embeddings, and cosine similarity.
- Ranked the top 5–10 candidates with similarity scores, fit verdicts, AI summaries, and downloadable review results.
- Evaluated **5 embedding models across 200+ resumes**, reaching **89% top-3 ranking accuracy**.

High-Performance Transformer Inference on Commodity GPUs GitHub
- Built a reproducible PyTorch benchmark across **6 model workloads** for latency, throughput, VRAM, OOM limits, and quality.
- Evaluated FlashAttention-2, SDPA, NF4, and INT4 KV-cache configurations on an RTX 3050 Ti with 4 GB VRAM.
- Increased maximum context from **6,784 to 8,064 tokens (+18.9%)** while keeping accuracy within 2 points of baseline.

ChatPDF Retrieval-Augmented Generation GitHub
- Built a Streamlit RAG application with PDF ingestion, chunking, embeddings, **FAISS vector search**, and retrieval.
- Integrated Bedrock LLM inference and repeatable FAISS index updates for document-grounded question answering.
- Added repeatable index refreshes and retrieval-context checks so answers used the latest indexed documents.

EDUCATION

San Jose State University, San Jose **GPA: 3.71/4**
MS, Electrical Engineering; Concentration in Artificial Intelligence Aug 2024 – May 2026
Relevant Coursework: Machine Learning, Neural Networks, Reinforcement Learning, Computer Vision, AI for DSP
Institute of Infrastructure Technology Research and Management, Ahmedabad
BTech, Electrical Engineering Aug 2017 – May 2021

RESEARCH PUBLICATIONS

- Sleep disorders identification via wavelet-based EOG and EMG features (second author), 2022. https://doi.org/10.1016/j.compbiomed.2022.105224
- Handling missing data in time-series data from wearables (first author), 2023. https://doi.org/10.5772/intechopen.1002536
- Efficient use of binned data for imputing univariate time-series data (first author), 2024. https://doi.org/10.3389/fdata.2024.1422650